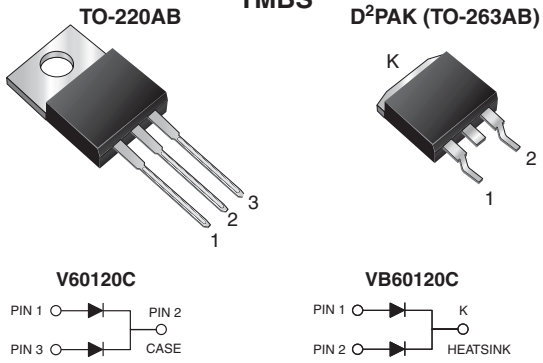
Dual High Voltage Trench MOS Barrier Schottky Rectifier

Ultra Low V_F = 0.41 V at I_F = 5 A

TMBS®



DESIGN SUPPORT TOOLS

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3D Models Available

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	2 x 30 A
V_{RRM}	120 V
I_{FSM}	300 A
V_F at I_F = 30 A	0.71 V
T_J max.	150 °C
Package	TO-220AB, D²PAK (TO-263AB)
Circuit configuration	Common cathode

FEATURES

- Trench MOS Schottky technology
- Low forward voltage drop, low power losses
- High efficiency operation
- Low thermal resistance
- Meets MSL level 1, per J-STD-020, LF maximum peak of 245 °C (for TO-263AB package)
- Solder bath temperature 275 °C maximum, 10 s, per JESD 22-B106 (for TO-220AB package)
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Pb e3 RoHS COMPLIANT

TYPICAL APPLICATIONS

For use in high frequency converters, switching power supplies, freewheeling diodes, OR-ing diode, DC/DC converters and reverse battery protection.

MECHANICAL DATA

Case: TO-220AB and D²PAK (TO-263AB)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
E3 suffix meets JESD 201 class 1A whisker test

Polarity: as marked

Mounting Torque: 10 in-lbs maximum

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	V60120C	VB60120C	UNIT
Maximum repetitive peak reverse voltage		V_{RRM}	120		V
Maximum average forward rectified current (fig. 1)	per device	$I_{F(AV)}$	60		A
	per diode		30		
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load per diode		I_{FSM}	300		A
Non-repetitive avalanche energy at T_J = 25 °C, L = 100 mH per diode		E_{AS}	260		mJ
Peak repetitive reverse current at t_p = 2 µs, 1 kHz, T_J = 38 °C ± 2 °C per diode		I_{RRM}	0.5		A
Voltage rate of change (rated V_R)		dV/dt	10 000		V/µs
Operating junction and storage temperature range		T_J, T_{STG}	-40 to +150		°C

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Breakdown voltage	I_R = 1.0 mA	T_A = 25 °C	V_{BR}	120 (minimum)	-	V
Instantaneous forward voltage per diode	I_F = 5 A	T_A = 25 °C	V_F (1)	0.48	-	V
	I_F = 15 A			0.66	-	
	I_F = 30 A			0.88	0.95	
	I_F = 5 A	T_A = 125 °C		0.41	-	
	I_F = 15 A			0.58	-	
	I_F = 30 A			0.71	0.75	
Reverse current at rated V_R per diode	V_R = 90 V	T_A = 25 °C	I_R (2)	14	-	µA
		T_A = 125 °C		11	-	mA
	V_R = 120 V	T_A = 25 °C		40	500	µA
		T_A = 125 °C		15	45	mA

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)				
PARAMETER	**SYMBOL**	**V60120C**	**VB60120C**	**UNIT**
Typical thermal resistance per diode	$R_{\theta JC}$	2.2	2.2	°C/W

ORDERING INFORMATION (Example)					
PACKAGE	**PREFERRED P/N**	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
TO-220AB	V60120C-E3/4W	1.89	4W	50/tube	Tube
TO-263AB	VB60120C-E3/4W	1.38	4W	50/tube	Tube
TO-263AB	VB60120C-E3/8W	1.38	8W	800/reel	Tape and reel

RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

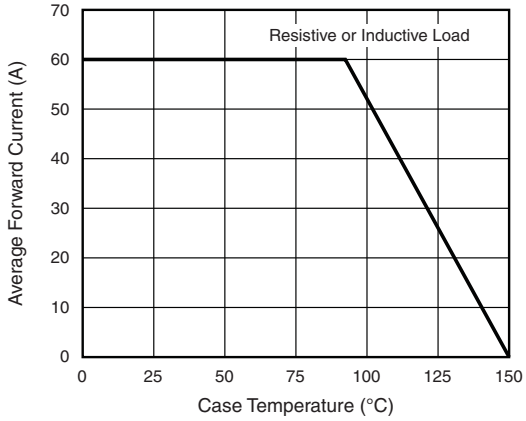


Fig. 1 - Forward Current Derating Curve

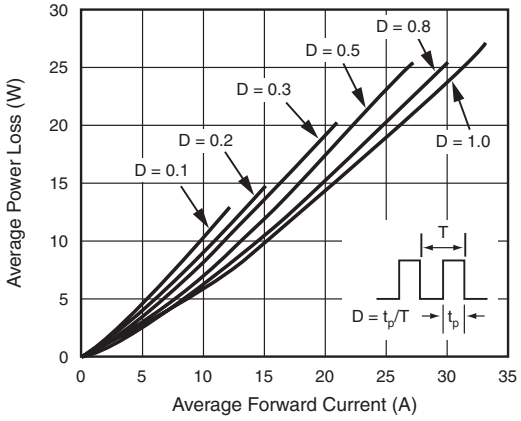


Fig. 2 - Forward Power Loss Characteristics Per Diode

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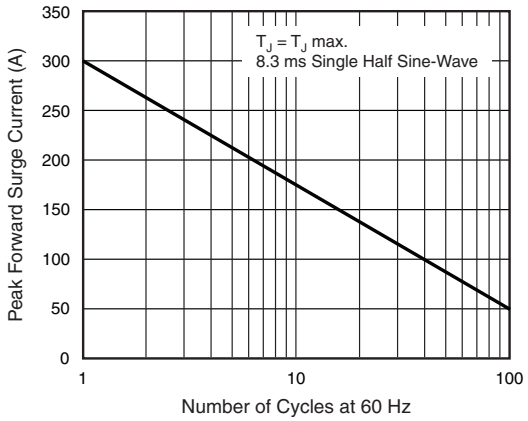


Fig. 3 - Maximum Non-Repetitive Peak Forward Surge Current Per Diode

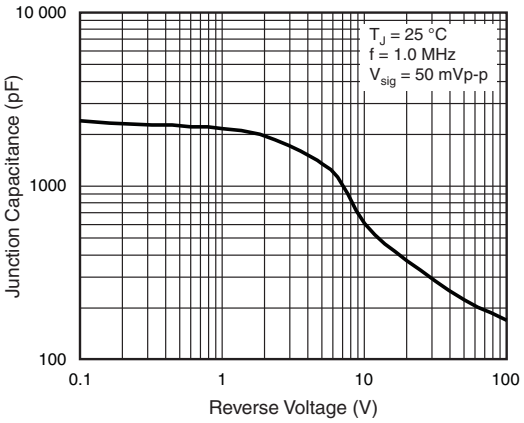


Fig. 6 - Typical Junction Capacitance Per Diode

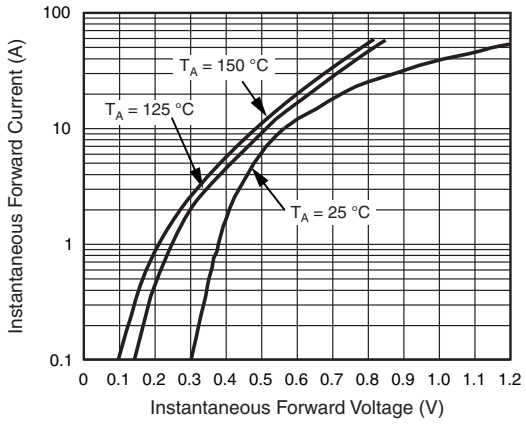


Fig. 4 - Typical Instantaneous Forward Characteristics Per Diode

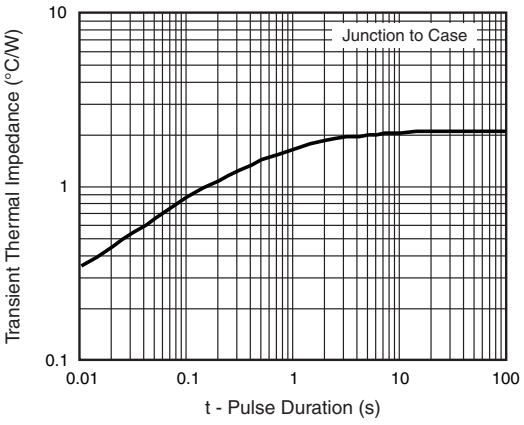


Fig. 7 - Typical Transient Thermal Impedance Per Diode

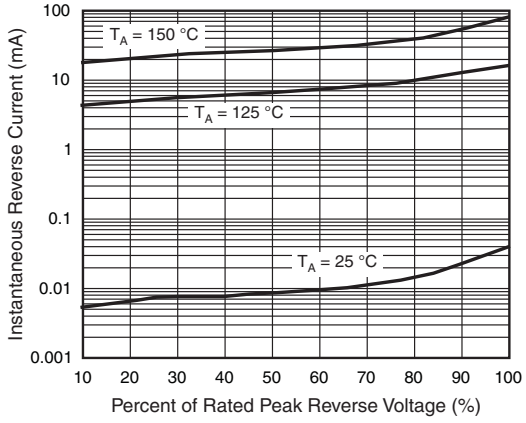
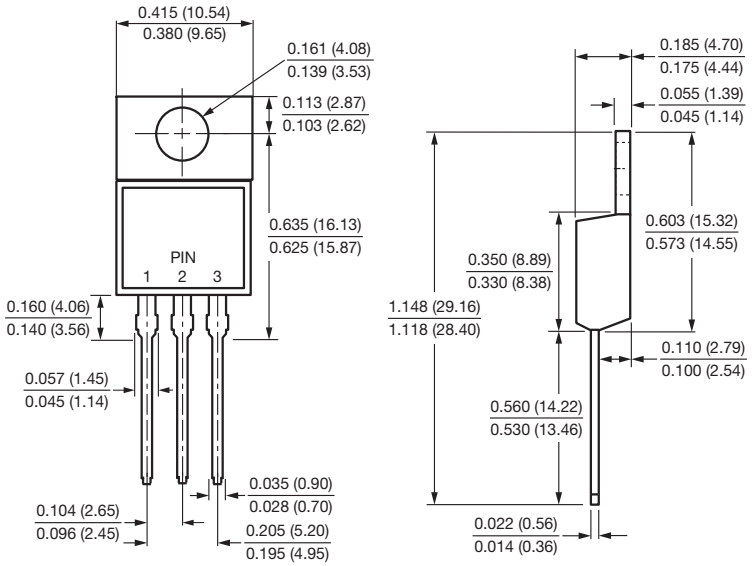


Fig. 5 - Typical Reverse Characteristics Per Diode

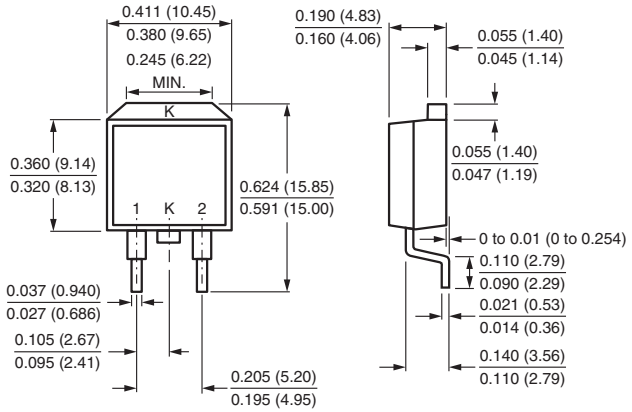
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

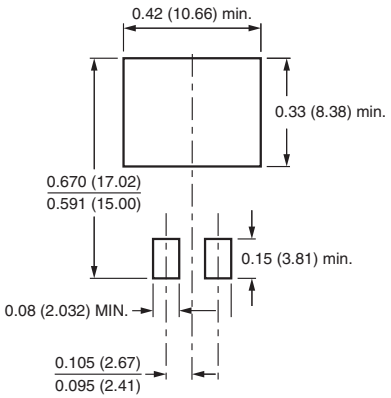
TO-220AB



D²PAK (TO-263AB)



Mounting Pad Layout



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